Truecaller: Scaling and launching new services quickly with Google Cloud

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About Truecaller

Developed by Swedish business True Software Scandinavia AB, Truecaller has built a communication app that is used by over 130 million users on a daily basis. The app provides a suite of unique services, such as a dialer that offers caller ID, spam detection, messaging, and financial services.

Industries: Technology

Google Cloud Results

- Enables launch of new highly secure financial communication service
- Delivers greater agility and speed-to-delivery of new products
- Completes migration in 63 days to minimize disruption to product development

Supp com app more 130M day

Location: India, Sweden

Cloud Storage

Google Cloud (https://cloud.google.com/)

Kubernetes Engine (https://cloud.google.com/kubernetes-engine/)

Cloud Dataproc (https://cloud.google.com/dataproc/)

Cloud Load Balancing (https://cloud.google.com/load-balancing/)

Cloud Storage (https://cloud.google.com/storage/)

Operations (https://cloud.google.com/products/operations)

Cloud Identity (https://cloud.google.com/identity/)

Kubernetes (https://cloud.google.com/kubernetes/)

Google Workspace (https://workspace.google.com/)

In India – and other countries – growing mobile connectivity and accessibility is creating demand for safe, efficient communications. Truecaller meets this need with a mobile application that helps people see who's trying to get in touch via phone calls or SMS, and offers an integrated caller ID service to enable users to block calls from unwanted parties. Truecaller – developed by Scandinavian business True Software Scandinavia AB – relies heavily on data to update its database of numbers and details.

"What stood out about Google Cloud Platform was its HTTP load balancing and managed Kubernetes, to automate the deployment, scaling, and management of containerized applications, through Google Kubernetes Engine."

—Pierre de Boer, Chief Operating Officer, Truecaller

"Our founders were second-generation immigrants to Sweden who wanted to simplify their lives," says Pierre de Boer, Chief Operating Officer, Truecaller. "They had a lot of friends abroad, but had no way of identifying who was calling them from overseas. They created a database of phone numbers and names that enabled them to see who was making a call."

Identifying callers

The number of names and numbers in the database expanded rapidly to a point Truecaller could use it to offer a comprehensive caller ID service. The founders then developed and marketed a mobile application that uses the stored data to enable users to identify callers. They also identified a second use case for subscribers – the ability to block calls as required.

"Our service is particularly valuable for women in India, some of whom may be harassed daily with unwanted calls and messages," says de Boer. "Truecaller enables them to identify unwanted callers and block them."

Despite the fact its parent business is headquartered in Sweden, about 65 percent to 70 percnent of Truecaller's business is in India, where its application is the fourth most downloaded and used – ahead of some global social media services.

Growth outstripped infrastructure

As Truecaller expanded, its infrastructure – housed in a data center in Sweden – was unable to support growth. "We needed to move as the business was maturing and we needed greater scale, flexibility, ease of deployment for our development pipelines, and longer term, access to AI and machine learning services," says de Boer.

Truecaller's leadership and technology teams realized these requirements could only be met by a cloud provider and evaluated the services available. The teams quickly realized Google Cloud (https://cloud.google.com/) was the best fit for its rest-of-world operations. "What stood out about Google Cloud Platform (https://cloud.google.com/) was its HTTP load balancing and managed Kubernetes (https://cloud.google.com/kubernetes/), to automate the deployment, scaling, and management of containerized applications, through Google Kubernetes Engine (https://cloud.google.com/kubernetes-engine/)," says de Boer. "Overall, we realized Google Cloud Platform would enable us to deploy faster and operate more easily." The business also established metrics in areas such as latency and deployment times that Google Cloud Platform could help it meet.

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Technical Account Manager to work with us for the entire journey."

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Four key workloads

Truecaller identified four key workloads as key to the success of the migration: its core infrastructure with 70 microservices; its content delivery network; its payment product; and a new messaging product. To meet deadlines for the launch of new products and to minimize time spent by team members, Truecaller set a tight timeline for the project. "We were then 110 people in the entire company and we were dedicating 50 people to this work, which meant we weren't delivering any other outputs," says de Boer. "By speeding this up, we could release resources faster to perform other work."

The business engaged Google Cloud Professional Services (https://cloud.google.com/consulting/) at the negotiation and proposal stage of the project to ensure a fast, smooth execution. "The Google Cloud Professional Services team spent a lot of time with us to ensure our success and assigned a Technical Account Manager to work with us for the entire journey," says De Boer. "With the Professional Service team's guidance, we were able to complete the migration quickly and efficiently, minimizing disruption to our product teams. We are continuing to work with the team on our longer-term cloud journey and we are extremely confident in their ability to help us realize our ambitions.

"Google Cloud Platform infrastructure provides a more efficient and faster way of configuring and setting up our compute instances than we have experienced in other environments," he adds. Truecaller's Google Cloud Platform architecture incorporates Google Kubernetes Engine to manage its containerized applications and Cloud Dataproc (https://cloud.google.com/dataproc/) to run its database clusters, while Stackdriver (https://cloud.google.com/stackdriver/) manages logs and analyzes operations.

Cloud Load Balancing (https://cloud.google.com/load-balancing/) enables the business to distribute load balanced compute resources to scale effectively, and Cloud Storage (https://cloud.google.com/storage/) provides highly secure, reliable online file storage.

Truecaller has also migrated from a cloud workplace applications solution to Google Workspace (https://workspace.google.com/solutions/). "This has enabled us to become significantly more collaborative and efficient internally," says De Boer. "Google Workspace brings so many advantages when it comes to fast, agile communication and cohesive working."

The business also uses Cloud Identity (https://cloud.google.com/identity/) endpoint administration and identity services. Integrated with Truecaller's internal directory service, Cloud Identity helps the business easily provision and manage the required user accounts for all its Google products.

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>
> *—Pierre de Boer, Chief Operating Officer, Truecaller*

"We also feel the advantage of testing out new products in alpha, which enables us to innovate in our infrastructure more than before," says De Boer. "Some of the products have been crucial for our setup, like HTTP/2 + gRPC to backend instances for HTTP(S) Load Balancer."

Broadening horizons

With over 130 million daily active users, Truecaller is broadening its horizons. "While caller ID and blocking is still the key use for our application, our vision and mission is now to make communications – transactions, messages, and calls – safe and efficient for our users," says de Boer. Truecaller aims to continue growing its presence in India as well as other high-growth markets.

Truecaller has introduced payments and started securing transactions between bank accounts operated by residents of India.

"Google Cloud Platform has exceeded most of our expectations – because we use autoscaling functionality for example, we don't even have to think about scalability anymore," de Boer says. Truecaller has reorganized from a matrix-based engineering product operation to a business unit oriented organization to take advantage of the agility and speed-to-deliver Google Cloud Platform enables.

With Google Cloud Platform, Truecaller is now ideally placed to move into the next stage of its development. "For the last decade we have acted as a startup and now we are moving to become a more mature company," says de Boer. "With Google Cloud, we will enable our developers to become part of a really exciting journey to change how we develop products and realize value."

Learn More

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(https://cloud.google.com